

Lossless Media

$$\mathbf{E}^i = \hat{\mathbf{y}} E_0^i e^{jk_1 z}, \quad \mathbf{H}^i = \hat{\mathbf{x}} \frac{E_0^i}{\eta_1} e^{jk_1 z}$$

$$\mathbf{E}^r = \hat{\mathbf{y}} \rho E_0^i e^{-jk_1 z}, \quad \mathbf{H}^r = -\hat{\mathbf{x}} \rho \frac{E_0^i}{\eta_1} e^{-jk_1 z}$$

$$\mathbf{E}^t = \hat{\mathbf{y}} \tau E_0^i e^{jk_2 z}, \quad \mathbf{H}^t = \hat{\mathbf{x}} \tau \frac{E_0^i}{\eta_2} e^{jk_2 z}$$

$$\text{Phase matching: } (\mathbf{E}^i + \mathbf{E}^r)|_{z=0} = \mathbf{E}^t|_{z=0}$$

$$(\mathbf{H}^i + \mathbf{H}^r)|_{z=0} = \mathbf{H}^t|_{z=0}$$

$$\rho = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau = 1 + \rho$$

$$k_1 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_1}, \quad k_2 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_2}$$

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon'_1}}, \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon'_2}}$$

Lossy Media

$$\mathbf{E}^i = \hat{\mathbf{y}} E_0^i e^{\gamma_1 z}, \quad \mathbf{H}^i = \hat{\mathbf{x}} \frac{E_0^i}{\eta_{c1}} e^{\gamma_1 z}$$

$$\mathbf{E}^r = \hat{\mathbf{y}} \rho E_0^i e^{-\gamma_1 z}, \quad \mathbf{H}^r = -\hat{\mathbf{x}} \rho \frac{E_0^i}{\eta_{c1}} e^{-\gamma_1 z}$$

$$\mathbf{E}^t = \hat{\mathbf{y}} \tau E_0^i e^{\gamma_2 z}, \quad \mathbf{H}^t = \hat{\mathbf{x}} \tau \frac{E_0^i}{\eta_{c2}} e^{\gamma_2 z}$$

$$\text{Phase matching: } (\mathbf{E}^i + \mathbf{E}^r)|_{z=0} = \mathbf{E}^t|_{z=0}$$

$$(\mathbf{H}^i + \mathbf{H}^r)|_{z=0} = \mathbf{H}^t|_{z=0}$$

$$\rho = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}}, \quad \tau = 1 + \rho$$

$$\gamma_1 = \alpha_1 + j\beta_1, \quad \gamma_2 = \alpha_2 + j\beta_2$$

$$\eta_{c1} = \frac{\eta_0}{\sqrt{\epsilon_1}}, \quad \eta_{c2} = \frac{\eta_0}{\sqrt{\epsilon_2}}$$

Notes: $\epsilon_1 = \epsilon'_1 - j\epsilon''_1$; $\epsilon_2 = \epsilon'_2 - j\epsilon''_2$.